

Campus Water & Leak Audit Report 2025

EXECUTIVE SUMMARY

This water conservation audit report outlines water intake, consumption intensity, and leakage identification across the residential hostels, dining blocks, and laboratories. The main goal is to improve water stewardship, lower municipal water bills, and establish alignment with UN SDG 6 (Clean Water and Sanitation).

1. KEY FOOTPRINT METRICS

Total annual campus water consumption stands at 9,85,50,000 liters (98.55 million liters). For a population of 2,000 residential students, this represents an average water usage intensity of 135 liters per capita per day (LPCD). The annual cost of raw water extraction and municipal sanitation supply charges totals INR 1,47,82,500.

2. IDENTIFIED INEFFICIENCIES & LEAKS

1. High Faucet Flow Rates: Faucets in the hostel restrooms deliver water at an average rate of 15 liters per minute. None are equipped with aerators or auto-cutoff mechanisms.
2. Overhead Tank Overflows: The main storage tanks suffer from faulty float valves. Four major overflow instances were reported this quarter, resulting in an estimated loss of 12,000 liters of potable water per day.
3. Underground Leakage: Pressure tests reveal a constant pressure drop between the main pump house and the sports field. An underground line fracture is suspected, causing a loss of about 15,000 liters daily.

3. ACTION PLAN & SDG CORRESPONDENCE

- Install 2.5 GPM low-flow aerators on all 350 public faucets. This simple retrofitting will cut tap water consumption by 50%.
- Replace all faulty mechanical float valves in the overhead reservoirs with electronic level controllers to eliminate overflow waste completely.
- Locate and repair the underground line fracture. Implementing sub-meters at each hostel block will help isolate future underground leaks.
- Restore and connect the rainwater harvesting sumps to the recharge wells. Roof runoff potential on campus is estimated at 4.2 million liters annually.
- Accomplishing these water efficiency goals will save 24.6 million liters of water annually, improving the score and directly meeting SDG 6 indicators.